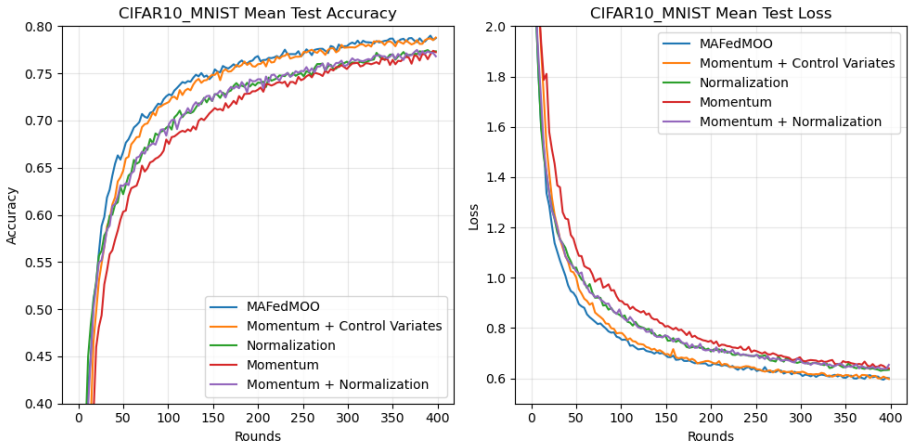

Figure 1: Ablation study on CIFAR10+MNIST dataset

Table 1: Comparisons of FSMGDA, FedCMOO and MAFedMOO

Method	**Key Assumptions**	**Conv. Rate**	**Comm. Cost**	**P-free**	**Mom.**
FSMGDA	Smoothness, Bounded Grad	$\mathcal{O}(1/T^{1/4})$	$\mathcal{O}(Md)$	×	×
FedCMOO	Smoothness, Bounded Var/Grad, Bounded Heterogeneity	$\mathcal{O}(1/T^{1/4})$	$\mathcal{O}(d)$	×	×
MAFedMOO	Smoothness, Bounded Var/Grad,	$\mathcal{O}(1/T^{1/4})$	$\mathcal{O}(Md)$	✓	✓

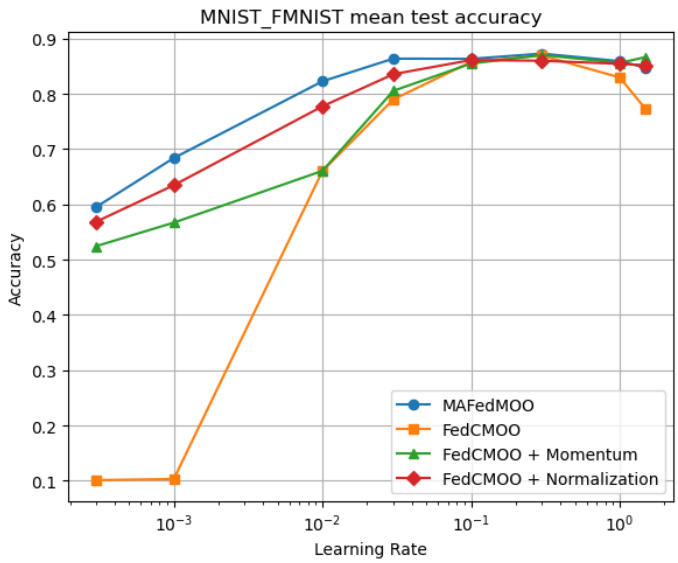

Figure 2: Mean test accuracy versus the learning rate on MNIST+FMNIST dataset

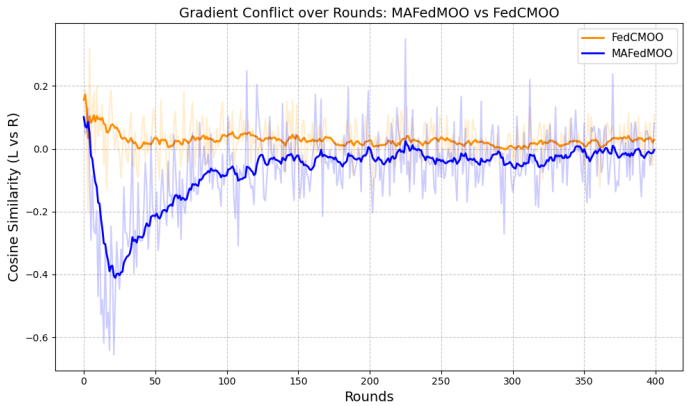

Figure 3: Conflict visualization on MNIST+FMNIST dataset